Malliavin calculus and Monte Carlo methods : Project E

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1 Part A : Some representation theorem

1.1 Question 1

Let

$$\mathcal{H} := \left\{ c + \int_0^T u_s \, dB_s \;:\; c \in \mathbb{R},\; u \in L^2_{\text{prog}}(\Omega \times [0,T]) \right\}.$$

We want to prove that $\mathcal{H}$ is dense in $L^2(\Omega)$, where by the notation of the statement

$$L^2(\Omega) = L^2(\Omega, \mathcal{F}_T, \mu).$$

We'll use the *Martingale Representation Theorem*:

Martingale Representation Theorem. *If $(M_t)_{0 \le t \le T}$ is a square-integrable martingale with respect to the Brownian filtration $(\mathcal{F}_t)_{0 \le t \le T}$, then there exists a unique process*

$$u \in L^2_{\text{prog}}(\Omega \times [0,T])$$

such that

$$M_t = M_0 + \int_0^t u_s \, dB_s, \qquad 0 \le t \le T.$$

Let $F \in L^2(\Omega)$ be arbitrary. Since

$$L^2(\Omega) = L^2(\Omega, \mathcal{F}_T, \mu),$$

the random variable F is $\mathcal{F}_T$-measurable and satisfies

$$\mathbb{E}[F^2] < \infty.$$

$$M_t := \mathbb{E}[F \mid \mathcal{F}_t], \qquad 0 \le t \le T.$$

By the properties of conditional expectation, $(M_t)_{0 \le t \le T}$ is an $(\mathcal{F}_t)$-martingale.

Moreover, it is square-integrable. Indeed, by Jensen's inequality for conditional expectation,

$$|M_t|^2 = |\mathbb{E}[F \mid \mathcal{F}_t]|^2 \le \mathbb{E}[F^2 \mid \mathcal{F}_t].$$

Taking expectation on both sides yields

$$\mathbb{E}[M_t^2] \le \mathbb{E}[F^2] < \infty.$$

Therefore (M_t) is a square-integrable martingale.

We may now apply the *Martingale Representation Theorem*. Hence there exists a unique process

$$u \in L^2_{\text{prog}}(\Omega \times [0,T])$$

such that

$$M_t = M_0 + \int_0^t u_s \, dB_s, \qquad 0 \le t \le T.$$

We now identify M_0 and M_T.

First, $\mathcal{F}_0$ in the Brownian setting is

$$M_0 = \mathbb{E}[F \mid \mathcal{F}_0] = \mathbb{E}[F].$$

Second, because F is $\mathcal{F}_T$-measurable, we have

$$M_T = \mathbb{E}[F \mid \mathcal{F}_T] = F.$$

Evaluating the representation at time T, we obtain

$$F = M_T = M_0 + \int_0^T u_s \, dB_s = \mathbb{E}[F] + \int_0^T u_s \, dB_s.$$

Thus F is of the form

$$c + \int_0^T u_s \, dB_s$$

with

$$c = \mathbb{E}[F] \in \mathbb{R}, \qquad u \in L^2_{\text{prog}}(\Omega \times [0, T]).$$

Therefore

$$F \in \mathcal{H}.$$

Since $F \in L^2(\Omega)$ was arbitrary, we have proved that

$$L^2(\Omega) \subset \mathcal{H}.$$

Conversely, every element of $\mathcal{H}$ clearly belongs to $L^2(\Omega)$, because for

$$X = c + \int_0^T u_s \, dB_s, \qquad u \in L^2_{\text{prog}}(\Omega \times [0, T]),$$

the Itô integral is well-defined and square-integrable by the *Itô isometry*, namely

$$\mathbb{E}\left[\left(\int_0^T u_s \, dB_s\right)^2\right] = \mathbb{E}\left[\int_0^T u_s^2 \, ds\right] < \infty.$$

Hence $\mathcal{H} \subset L^2(\Omega)$.

We conclude that

$$\mathcal{H} = L^2(\Omega).$$

Since $\mathcal{H} = L^2(\Omega)$, it is in particular dense in $L^2(\Omega)$.

1.2 Question 2

2 a) Let

$$G := F - \mathbb{E}[F].$$

Since $F \in \mathbb{D}$ and $\mathbb{E}[F]$ is a deterministic constant, we still have

$$G \in \mathbb{D}.$$

Moreover,we have :

$$D_s G = D_s(F - \mathbb{E}[F]) = D_s F.$$

Now suppose that the desired representation formula has been proved for every centered random variable in $\mathbb{D}$. Since

$$\mathbb{E}[G] = \mathbb{E}[F] - \mathbb{E}[F] = 0,$$

we may apply it to G and obtain

$$G = \int_0^T \mathbb{E}[D_s G \mid \mathcal{F}_s]\, dB_s.$$

Using $G = F - \mathbb{E}[F]$ and $D_s G = D_s F$, this becomes

$$F - \mathbb{E}[F] = \int_0^T \mathbb{E}[D_s F \mid \mathcal{F}_s]\, dB_s.$$

Hence,

$$F = \mathbb{E}[F] + \int_0^T \mathbb{E}[D_s F \mid \mathcal{F}_s]\, dB_s.$$

Therefore, proving the formula for F is equivalent to proving it for the centered variable $G = F - \mathbb{E}[F]$. Thus, we may assume without loss of generality that

$$\mathbb{E}[F] = 0.$$

2 b) We will assume :

$$\mathbb{E}[F] = 0.$$

Let

$$v_s := \mathbb{E}[D_s F \mid \mathcal{F}_s], \qquad 0 \le s \le T.$$

So we have $v \in L^2_{\text{prog}}(\Omega \times [0,T])$. since $F \in \mathbb{D}$ and by definition of $\mathbb{D}$ we get

$$\mathbb{E}\left[\int_0^T |D_s F|^2\, ds\right] < \infty.$$

Moreover, by Jensen's inequality

$$|v_s|^2 = |\mathbb{E}[D_s F \mid \mathcal{F}_s]|^2 \le \mathbb{E}[|D_s F|^2 \mid \mathcal{F}_s].$$

From the former we get the following by integrating over $s \in [0,T]$

$$\mathbb{E}\left[\int_0^T |v_s|^2\, ds\right] \le \mathbb{E}\left[\int_0^T |D_s F|^2\, ds\right] < \infty.$$

Hence $v \in L^2_{\text{prog}}(\Omega \times [0,T])$.

Now let $c \in \mathbb{R}$ and $u \in L^2_{\text{prog}}(\Omega \times [0,T])$. We want to prove that

$$\mathbb{E}\left[F\left(c + \int_0^T u_s\, dB_s\right)\right] = \mathbb{E}\left[\left(\int_0^T v_s\, dB_s\right)\left(c + \int_0^T u_s\, dB_s\right)\right].$$

Since $\mathbb{E}[F]=0$, the constant term on the left-hand side vanishes:

$$\mathbb{E}\left[F\left(c+\int_0^T u_s\,dB_s\right)\right]=c\,\mathbb{E}[F]+\mathbb{E}\left[F\int_0^T u_s\,dB_s\right]=\mathbb{E}\left[F\int_0^T u_s\,dB_s\right].$$

Because $u\in L^2_{\text{prog}}(\Omega\times[0,T])$, the course gives

$$\delta(u)=\int_0^T u_s\,dB_s,$$

and by the relation between D and δ,

$$\mathbb{E}\left[F\int_0^T u_s\,dB_s\right]=\mathbb{E}[F\,\delta(u)]=\mathbb{E}\left[\int_0^T D_sF\,u_s\,ds\right].$$

Since u_s is $\mathcal{F}_s$-measurable, the tower property yields

$$\mathbb{E}[D_sF\,u_s]=\mathbb{E}\big[\mathbb{E}[D_sF\mid\mathcal{F}_s]\,u_s\big]=\mathbb{E}[v_su_s].$$

Therefore,

$$\mathbb{E}\left[F\left(c+\int_0^T u_s\,dB_s\right)\right]=\mathbb{E}\left[\int_0^T v_su_s\,ds\right].$$

We now compute the right-hand side. Since $v\in L^2_{\text{prog}}(\Omega\times[0,T])$, the Itô integral $\int_0^T v_s\,dB_s$ is well defined and centered, so

$$\mathbb{E}\left[\left(\int_0^T v_s\,dB_s\right)\left(c+\int_0^T u_s\,dB_s\right)\right]=c\,\mathbb{E}\left[\int_0^T v_s\,dB_s\right]+\mathbb{E}\left[\left(\int_0^T v_s\,dB_s\right)\left(\int_0^T u_s\,dB_s\right)\right].$$

The first term is equal to zero. For the second one, using Itô's isometry,

$$\mathbb{E}\left[\left(\int_0^T v_s\,dB_s\right)\left(\int_0^T u_s\,dB_s\right)\right]=\mathbb{E}\left[\int_0^T v_su_s\,ds\right].$$

Hence,

$$\mathbb{E}\left[\left(\int_0^T v_s\,dB_s\right)\left(c+\int_0^T u_s\,dB_s\right)\right]=\mathbb{E}\left[\int_0^T v_su_s\,ds\right].$$

Combining the two previous identities, we conclude that

$$\mathbb{E}\left[F\left(c+\int_0^T u_s\,dB_s\right)\right]=\mathbb{E}\left[\left(\int_0^T \mathbb{E}[D_sF\mid\mathcal{F}_s]\,dB_s\right)\left(c+\int_0^T u_s\,dB_s\right)\right].$$

2 c) We now prove that
$$F = \mathbb{E}[F] + \int_0^T \mathbb{E}[D_s F \mid \mathcal{F}_s]\, dB_s. \tag{1}$$
From question 2 a) , we can assume without loss of generality :
$$\mathbb{E}[F] = 0.$$
Therefore, we'll assume that F is centered.

Let
$$v_s := \mathbb{E}[D_s F \mid \mathcal{F}_s], \qquad 0 \le s \le T,$$
and define
$$G := F - \int_0^T v_s\, dB_s.$$
From question 2 b), for every $c \in \mathbb{R}$ and every $u \in L^2_{\text{prog}}(\Omega \times [0,T])$, we have
$$\mathbb{E}\left[F\left(c + \int_0^T u_s\, dB_s\right)\right] = \mathbb{E}\left[\left(\int_0^T v_s\, dB_s\right)\left(c + \int_0^T u_s\, dB_s\right)\right].$$
Subtracting the right-hand side from the left-hand side gives
$$\mathbb{E}\left[G\left(c + \int_0^T u_s\, dB_s\right)\right] = 0$$
for all $c \in \mathbb{R}$ and all $u \in L^2_{\text{prog}}(\Omega \times [0,T])$.

In other words, G is orthogonal in $L^2(\Omega)$ to every element of the set
$$\mathcal{H} = \left\{ c + \int_0^T u_s\, dB_s \;:\; c \in \mathbb{R},\; u \in L^2_{\text{prog}}(\Omega \times [0,T]) \right\}.$$
But from question 1, we showed that the set $\mathcal{H}$ is equal to $L^2(\Omega)$, so dense in it. The following mapping
$$X \longmapsto \mathbb{E}[GX]$$
is continuous on $L^2(\Omega)$. Therefore,
$$\mathbb{E}[GX] = 0, \qquad \forall X \in L^2(\Omega).$$
Taking in particular $X = G$, we obtain
$$\mathbb{E}[G^2] = 0.$$
Possible if and only if we have
$$G = 0 \qquad \text{in } L^2(\Omega),$$
Hence
$$F = \int_0^T \mathbb{E}[D_s F \mid \mathcal{F}_s]\, dB_s.$$
This proves (1) in the centered case. Since we don't lose the generality we have:
$$F = \mathbb{E}[F] + \int_0^T \mathbb{E}[D_s F \mid \mathcal{F}_s]\, dB_s.$$

2 d) Let $(M_t)_{0\le t\le T}$ be a square-integrable martingale with respect to $(\mathcal{F}_t)_{0\le t\le T}$ such that $M_T \in \mathbb{D}$. We claim that (M_t) admits the representation

$$M_t = \mathbb{E}[M_T] + \int_0^t \mathbb{E}[D_s M_T \mid \mathcal{F}_s]\, dB_s, \qquad 0 \le t \le T. \tag{2}$$

Proof : Since (M_t) is a square-integrable martingale, we have

$$M_t = \mathbb{E}[M_T \mid \mathcal{F}_t], \qquad 0 \le t \le T.$$

Moreover, $M_T \in \mathbb{D}$, so by the formula we proved in question 2 c),

$$M_T = \mathbb{E}[M_T] + \int_0^T \mathbb{E}[D_s M_T \mid \mathcal{F}_s]\, dB_s.$$

Set

$$u_s := \mathbb{E}[D_s M_T \mid \mathcal{F}_s], \qquad 0 \le s \le T.$$

Then $u \in L^2_{\text{prog}}(\Omega \times [0,T])$, so the Itô integral $\int_0^T u_s\, dB_s$ is well defined.

Taking conditional expectation with respect to $\mathcal{F}_t$, we obtain

$$M_t = \mathbb{E}[M_T \mid \mathcal{F}_t] = \mathbb{E}[M_T] + \mathbb{E}\left[\int_0^T u_s\, dB_s \,\middle|\, \mathcal{F}_t\right].$$

Now split the integral at time t:

$$\int_0^T u_s\, dB_s = \int_0^t u_s\, dB_s + \int_t^T u_s\, dB_s.$$

Since $\int_0^t u_s\, dB_s$ is $\mathcal{F}_t$-measurable,

$$\mathbb{E}\left[\int_0^t u_s\, dB_s \,\middle|\, \mathcal{F}_t\right] = \int_0^t u_s\, dB_s.$$

Also, because $\left(\int_0^r u_s\, dB_s\right)_{0\le r\le T}$ is a square-integrable martingale,

$$\mathbb{E}\left[\int_t^T u_s\, dB_s \,\middle|\, \mathcal{F}_t\right] = 0.$$

Therefore,

$$\mathbb{E}\left[\int_0^T u_s\, dB_s \,\middle|\, \mathcal{F}_t\right] = \int_0^t u_s\, dB_s.$$

Substituting this into the previous identity gives

$$M_t = \mathbb{E}[M_T] + \int_0^t u_s\, dB_s = \mathbb{E}[M_T] + \int_0^t \mathbb{E}[D_s M_T \mid \mathcal{F}_s]\, dB_s.$$

This proves (2). Since $M_0 = \mathbb{E}[M_T]$, the formula may also be written as

$$M_t = M_0 + \int_0^t \mathbb{E}[D_s M_T \mid \mathcal{F}_s]\, dB_s, \qquad 0 \le t \le T.$$

1.3 Question 3

3 a) Let us write

$$S_t := X_t^x, \qquad \text{so that} \qquad dS_t = rS_t\,dt + \sigma S_t\,dB_t, \qquad S_0 = x,$$

and let

$$S_t^0 := e^{rt}$$

denote the risk-free asset.

Assume that a self-financed portfolio $(H_t, \eta_t)_{0\le t\le T}$ replicates F, where H_t is the quantity of risky asset and η_t the quantity of risk-free asset. Its wealth is

$$V_t = H_t S_t + \eta_t S_t^0,$$

and the self-financing condition gives

$$dV_t = H_t\,dS_t + \eta_t\,dS_t^0.$$

Since

$$dS_t = rS_t\,dt + \sigma S_t\,dB_t \qquad \text{and} \qquad dS_t^0 = rS_t^0\,dt,$$

we obtain

$$dV_t = H_t(rS_t\,dt + \sigma S_t\,dB_t) + \eta_t r S_t^0\,dt = r(H_t S_t + \eta_t S_t^0)\,dt + \sigma S_t H_t\,dB_t.$$

Therefore

$$dV_t = rV_t\,dt + \sigma S_t H_t\,dB_t.$$

Now we introduce the discounted wealth which is

$$\widetilde{V}_t := e^{-rt} V_t.$$

The derivative is

$$d\widetilde{V}_t = d(e^{-rt}V_t) = e^{-rt}\,dV_t - re^{-rt}V_t\,dt.$$

Substituting the expression for dV_t found above,

$$d\widetilde{V}_t = e^{-rt}(rV_t\,dt + \sigma S_t H_t\,dB_t) - re^{-rt}V_t\,dt = e^{-rt}\sigma S_t H_t\,dB_t.$$

Hence

$$\widetilde{V}_T = V_0 + \int_0^T e^{-rs}\sigma S_s H_s\,dB_s.$$

Since the portfolio hedges F, we have $V_T = F$, so

$$e^{-rT}F = V_0 + \int_0^T e^{-rs}\sigma S_s H_s\,dB_s. \tag{3}$$

On the other hand, $F \in \mathbb{D}$ by assumption, and e^{-rT} is deterministic, so

$$e^{-rT}F \in \mathbb{D} \qquad \text{and} \qquad D_s(e^{-rT}F) = e^{-rT}D_sF.$$

Applying the representation formula proved in question 2 to $e^{-rT}F$, we get

$$e^{-rT}F = \mathbb{E}[e^{-rT}F] + \int_0^T \mathbb{E}[D_s(e^{-rT}F) \mid \mathcal{F}_s]\, dB_s,$$

that is,

$$e^{-rT}F = \mathbb{E}[e^{-rT}F] + \int_0^T e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s]\, dB_s. \tag{4}$$

Comparing (3) and (4), we can write the following equality

$$V_0 - \mathbb{E}[e^{-rT}F] = \int_0^T \left(e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s] - e^{-rs}\sigma S_s H_s\right)\, dB_s.$$

The left-hand side is deterministic, while the Itô integral on the right-hand side has mean zero. Hence necessarily

$$V_0 = \mathbb{E}[e^{-rT}F],$$

and therefore

$$\int_0^T \left(e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s] - e^{-rs}\sigma S_s H_s\right)\, dB_s = 0.$$

By Itô's isometry,

$$\begin{aligned} 0 &= \mathbb{E}\left[\left(\int_0^T \left(e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s] - e^{-rs}\sigma S_s H_s\right)\, dB_s\right)^2\right] \\ &= \mathbb{E}\left[\int_0^T \left(e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s] - e^{-rs}\sigma S_s H_s\right)^2 ds\right]. \end{aligned}$$

Thus

$$e^{-rs}\sigma S_s H_s = e^{-rT}\mathbb{E}[D_sF \mid \mathcal{F}_s] \qquad .$$

Equivalently,

$$H_s = \frac{\mathbb{E}[D_sF \mid \mathcal{F}_s]}{\sigma S_s e^{r(T-s)}}.$$

Renaming s as t, we finally obtain

$$H_t = \frac{\mathbb{E}[D_tF \mid \mathcal{F}_t]}{\sigma S_t e^{r(T-t)}}, \qquad 0 \le t \le T,$$

3 b) Assume now that $\Phi \in C^1(\mathbb{R}) \cap \mathrm{Lip}(\mathbb{R},\mathbb{R})$, and let

$$F = \Phi(S_T), \qquad \text{where } S_t := X_t^x.$$

We want to show that the strategy obtained in question 3 a) coincides with the usual delta hedging, namely

$$H_t = \partial_x v(t, S_t),$$

where the Black–Scholes price function is defined by

$$v(t,x) := e^{-r(T-t)}\mathbb{E}\big[\Phi(X_T^{t,x})\big].$$

Here $X^{t,x}$ denotes the solution of the Black–Scholes SDE started from x at time t, that is

$$X_T^{t,x} = x\exp\left(\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(B_T - B_t)\right).$$

Set

$$G_{t,T} := \exp\left(\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(B_T - B_t)\right),$$

so that

$$X_T^{t,x} = xG_{t,T}.$$

Hence

$$v(t,x) = e^{-r(T-t)}\mathbb{E}\big[\Phi(xG_{t,T})\big].$$

Since $\Phi \in C^1(\mathbb{R}) \cap \mathrm{Lip}(\mathbb{R},\mathbb{R})$, its derivative Φ' is bounded. Therefore we may differentiate under the expectation sign, and obtain the expression of the delta

$$\partial_x v(t,x) = e^{-r(T-t)}\mathbb{E}\big[\Phi'(xG_{t,T})G_{t,T}\big]. \tag{5}$$

Since

$$S_T = x\exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma B_T\right),$$

we have, for $0 \le t \le T$,

$$D_t S_T = \sigma S_T.$$

Moreover, by the chain rule we get

$$D_t F = D_t(\Phi(S_T)) = \Phi'(S_T)D_t S_T = \sigma S_T \Phi'(S_T). \tag{6}$$

From the previous question we have

$$H_t = \frac{\mathbb{E}[D_t F \mid \mathcal{F}_t]}{\sigma S_t e^{r(T-t)}}.$$

Using (6), this becomes

$$H_t = \frac{\mathbb{E}[\sigma S_T \Phi'(S_T) \mid \mathcal{F}_t]}{\sigma S_t e^{r(T-t)}} = e^{-r(T-t)}\frac{1}{S_t}\mathbb{E}[S_T\Phi'(S_T) \mid \mathcal{F}_t].$$

Now

$$S_T = S_t G_{t,T},$$

therefore

$$H_t = e^{-r(T-t)}\mathbb{E}\big[\Phi'(S_t G_{t,T})G_{t,T} \mid \mathcal{F}_t\big].$$

Since $G_{t,T}$ depends only on $B_T - B_t$, it is independent of $\mathcal{F}_t$. Thus

$$\mathbb{E}\big[\Phi'(S_t G_{t,T})G_{t,T} \mid \mathcal{F}_t\big] = \psi_t(S_t),$$

where

$$\psi_t(x) := \mathbb{E}\big[\Phi'(xG_{t,T})G_{t,T}\big].$$

Hence

$$H_t = e^{-r(T-t)}\mathbb{E}\big[\Phi'(S_tG_{t,T})G_{t,T}\big].$$

Using (5), we conclude that

$$H_t = \partial_x v(t, S_t).$$

Therefore the strategy obtained in question 3 a) is exactly the usual delta hedging strategy.

2 Part B : Complex Asian options in the Bachelier Model

2.1 Question a)

Let

$$A(x) := \int_0^T X_s^x\,ds, \qquad B(x) := X_T^x,$$

so that

$$P(x) = e^{-rT}\,\mathbb{E}\big[\Phi(A(x), B(x))\big].$$

We use Proposition 8 from the course: under the stated assumptions on the SDE, for every $t \in [0, T]$, the map $x \mapsto X_t^x$ is differentiable and

$$\frac{\partial X_t^x}{\partial x} = Y_t^x,$$

where Y^x is the first variation process.

We now prove the formula for $P'(x)$.

Let $h \neq 0$. We write

$$A_h := A(x+h) = \int_0^T X_s^{x+h}\,ds, \qquad B_h := B(x+h) = X_T^{x+h},$$

and

$$A := A(x) = \int_0^T X_s^x\,ds, \qquad B := B(x) = X_T^x.$$

Then

$$\frac{P(x+h) - P(x)}{h} = e^{-rT}\,\mathbb{E}\left[\frac{\Phi(A_h, B_h) - \Phi(A, B)}{h}\right].$$

Since $\Phi \in C^1(\mathbb{R}^2)$, we use the mean value formula in integral form:

$$\Phi(A_h, B_h) - \Phi(A, B) = \int_0^1 \partial_{x_1}\Phi\big(A + \theta(A_h - A),\, B + \theta(B_h - B)\big)\,(A_h - A)\,d\theta$$
$$+ \int_0^1 \partial_{x_2}\Phi\big(A + \theta(A_h - A),\, B + \theta(B_h - B)\big)\,(B_h - B)\,d\theta.$$

Hence

$$\frac{\Phi(A_h,B_h)-\Phi(A,B)}{h} = \int_0^1 \partial_{x_1}\Phi\big(A+\theta(A_h-A),\, B+\theta(B_h-B)\big)\,\frac{A_h-A}{h}\,d\theta$$
$$+\int_0^1 \partial_{x_2}\Phi\big(A+\theta(A_h-A),\, B+\theta(B_h-B)\big)\,\frac{B_h-B}{h}\,d\theta. \tag{7}$$

Now, by Proposition 8,

$$\frac{X_s^{x+h}-X_s^x}{h} \longrightarrow Y_s^x \quad \text{in } L^2(\Omega) \qquad \text{for every } s\in[0,T],$$

and

$$\frac{X_T^{x+h}-X_T^x}{h} \longrightarrow Y_T^x \quad \text{in } L^2(\Omega).$$

Therefore, by Cauchy–Schwarz,

$$\frac{A_h-A}{h} = \int_0^T \frac{X_s^{x+h}-X_s^x}{h}\,ds \longrightarrow \int_0^T Y_s^x\,ds \quad \text{in } L^2(\Omega),$$

since

$$\mathbb{E}\left[\left|\int_0^T \left(\frac{X_s^{x+h}-X_s^x}{h}-Y_s^x\right)ds\right|^2\right] \le T\int_0^T \mathbb{E}\left[\left|\frac{X_s^{x+h}-X_s^x}{h}-Y_s^x\right|^2\right]ds.$$

Similarly,

$$\frac{B_h-B}{h} \longrightarrow Y_T^x \quad \text{in } L^2(\Omega).$$

Moreover,

$$A_h \longrightarrow A, \qquad B_h \longrightarrow B \quad \text{in } L^2(\Omega),$$

hence also in probability. Since $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2,\mathbb{R})$, its partial derivatives are bounded and continuous. Therefore, for every $\theta\in[0,1]$,

$$\partial_{x_i}\Phi\big(A+\theta(A_h-A),\, B+\theta(B_h-B)\big) \longrightarrow \partial_{x_i}\Phi(A,B) \quad \text{in probability,}$$

and the convergence is dominated by a constant.

Passing to the limit in (7) by dominated convergence, we obtain

$$\frac{P(x+h)-P(x)}{h} \longrightarrow e^{-rT}\,\mathbb{E}\left[\partial_{x_1}\Phi(A,B)\int_0^T Y_s^x\,ds + \partial_{x_2}\Phi(A,B)\,Y_T^x\right].$$

Thus P is differentiable and

$$P'(x) = e^{-rT}\,\mathbb{E}\left[\partial_{x_1}\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)\int_0^T Y_s^x\,ds\right] + e^{-rT}\,\mathbb{E}\left[\partial_{x_2}\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)Y_T^x\right].$$

In other words,

$$\Delta(x) = \frac{dP(x)}{dx} = e^{-rT}\,\mathbb{E}\left[\partial_{x_1}\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)\int_0^T Y_s^x\,ds\right] + e^{-rT}\,\mathbb{E}\left[\partial_{x_2}\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)Y_T^x\right].$$

Finally, the same argument together with the continuity in x of (X^x, Y^x) given by Proposition 8 shows that $x\mapsto P'(x)$ is continuous. Hence

$$P\in C^1(\mathbb{R}).$$

2.2 Question b)

Let

$$A := \int_0^T X_s^x \, ds, \qquad B := X_T^x, \qquad \mathcal{G} := \sigma(A, B),$$

so that for every $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$,

$$\Phi\left(\int_0^T X_s^x \, ds, X_T^x\right) = \Phi(A, B)$$

is $\mathcal{G}$-measurable.

Let $\Pi \in W$ be arbitrary, and define

$$\Pi_0 := \mathbb{E}[\Pi \mid \mathcal{G}].$$

We prove that $\Pi_0 \in W$ and that it minimizes the variance of

$$\Phi(A, B)\Pi$$

among all $\Pi \in W$.

Since $\Pi \in L^2(\Omega)$ and conditional expectation is a contraction in L^2,

$$\mathbb{E}[\Pi_0^2] = \mathbb{E}\big[\big(\mathbb{E}[\Pi \mid \mathcal{G}]\big)^2\big] \leq \mathbb{E}[\Pi^2] < \infty.$$

Hence

$$\Pi_0 \in L^2(\Omega).$$

Fix $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$ and set

$$Z := \Phi(A, B).$$

Since Z is $\mathcal{G}$-measurable, we have

$$\mathbb{E}[Z\Pi_0] = \mathbb{E}\big[Z \, \mathbb{E}[\Pi \mid \mathcal{G}]\big] = \mathbb{E}\big[\mathbb{E}[Z\Pi \mid \mathcal{G}]\big] = \mathbb{E}[Z\Pi].$$

Because $\Pi \in W$, by definition of W,

$$\Delta(x) = e^{-rT}\mathbb{E}[Z\Pi].$$

Therefore,

$$\Delta(x) = e^{-rT}\mathbb{E}[Z\Pi_0].$$

Since this holds for every $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$, we conclude that

$$\Pi_0 \in W.$$

Again let

$$Z := \Phi(A, B).$$

Since Z is $\mathcal{G}$-measurable,

$$\mathbb{E}[Z\Pi \mid \mathcal{G}] = Z \, \mathbb{E}[\Pi \mid \mathcal{G}] = Z\Pi_0.$$

Now apply the law of total variance to the square-integrable random variable $Z\Pi$:

$$\mathrm{Var}(Z\Pi) = \mathrm{Var}\big(\mathbb{E}[Z\Pi \mid \mathcal{G}]\big) + \mathbb{E}\big[\mathrm{Var}(Z\Pi \mid \mathcal{G})\big].$$

Using the previous identity, this becomes

$$\mathrm{Var}(Z\Pi) = \mathrm{Var}(Z\Pi_0) + \mathbb{E}\big[\mathrm{Var}(Z\Pi \mid \mathcal{G})\big] \geq \mathrm{Var}(Z\Pi_0).$$

Hence

$$\mathrm{Var}\left(\Phi\left(\int_0^T X_s^x\, ds, X_T^x\right)\Pi_0\right) \leq \mathrm{Var}\left(\Phi\left(\int_0^T X_s^x\, ds, X_T^x\right)\Pi\right)$$

for every $\Pi \in W$.

Therefore, the minimum-variance weight is given by

$$\Pi_0 = \mathbb{E}\left[\Pi \,\middle|\, \sigma\left(\int_0^T X_s^x\, ds, X_T^x\right)\right],$$

where Π is any arbitrary element of W.

2.3 Question c)

Let

$$A := \int_0^T X_s^x\, ds, \qquad B := X_T^x, \qquad \mathcal{G} := \sigma(A, B), \qquad \Pi = \delta(w),$$

with $w \in \mathrm{dom}(\delta)$.

We prove that

$$\Delta(x) = e^{-rT}\,\mathbb{E}\big[\Phi(A,B)\Pi\big], \qquad \forall \Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R}), \tag{8}$$

if and only if

$$\mathbb{E}\left[\int_0^T Y_t^x \int_0^t \frac{\sigma(s, X_s^x)}{Y_s^x} w_s\, ds\, dt \,\middle|\, \mathcal{G}\right] = \mathbb{E}\left[\int_0^T Y_t^x\, dt \,\middle|\, \mathcal{G}\right], \tag{9}$$

and

$$\mathbb{E}\left[Y_T^x \int_0^T \frac{\sigma(s, X_s^x)}{Y_s^x} w_s\, ds \,\middle|\, \mathcal{G}\right] = \mathbb{E}\left[Y_T^x \mid \mathcal{G}\right]. \tag{10}$$

Firsy with proposition 8 of the course, for $0 \leq s \leq t \leq T$, we have

$$D_s X_t^x = Y_t^x \frac{\sigma(s, X_s^x)}{Y_s^x}.$$

Hence

$$D_s A = D_s\left(\int_0^T X_t^x\, dt\right) = \int_s^T D_s X_t^x\, dt = \frac{\sigma(s, X_s^x)}{Y_s^x}\int_s^T Y_t^x\, dt,$$

and

$$D_s B = D_s X_T^x = Y_T^x \frac{\sigma(s, X_s^x)}{Y_s^x}.$$

Now let $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$. By the chain rule,

$$D_s\Phi(A,B) = \partial_{x_1}\Phi(A,B)\, D_sA + \partial_{x_2}\Phi(A,B)\, D_sB.$$

Therefore

$$D_s\Phi(A,B) = \partial_{x_1}\Phi(A,B)\, \frac{\sigma(s,X_s^x)}{Y_s^x} \int_s^T Y_t^x\, dt + \partial_{x_2}\Phi(A,B)\, Y_T^x \frac{\sigma(s,X_s^x)}{Y_s^x}.$$

Since $\Pi = \delta(w)$, the duality relation between D and δ gives

$$\mathbb{E}[\Phi(A,B)\Pi] = \mathbb{E}\left[\int_0^T D_s\Phi(A,B)\, w_s\, ds\right].$$

Thus

$$\begin{aligned}\mathbb{E}[\Phi(A,B)\Pi] = \mathbb{E}&\left[\partial_{x_1}\Phi(A,B) \int_0^T \frac{\sigma(s,X_s^x)}{Y_s^x}\left(\int_s^T Y_t^x\, dt\right) w_s\, ds\right] \\ &+\mathbb{E}\left[\partial_{x_2}\Phi(A,B)\, Y_T^x \int_0^T \frac{\sigma(s,X_s^x)}{Y_s^x} w_s\, ds\right].\end{aligned}$$

Using Fubini's theorem in the first term,

$$\int_0^T \frac{\sigma(s,X_s^x)}{Y_s^x}\left(\int_s^T Y_t^x\, dt\right) w_s\, ds = \int_0^T Y_t^x \int_0^t \frac{\sigma(s,X_s^x)}{Y_s^x} w_s\, ds\, dt.$$

Hence

$$\mathbb{E}[\Phi(A,B)\Pi] = \mathbb{E}\left[\partial_{x_1}\Phi(A,B)\, U\right] + \mathbb{E}\left[\partial_{x_2}\Phi(A,B)\, V\right], \tag{11}$$

where

$$U := \int_0^T Y_t^x \int_0^t \frac{\sigma(s,X_s^x)}{Y_s^x} w_s\, ds\, dt, \qquad V := Y_T^x \int_0^T \frac{\sigma(s,X_s^x)}{Y_s^x} w_s\, ds.$$

From question a), we know that

$$\Delta(x) = e^{-rT}\, \mathbb{E}\left[\partial_{x_1}\Phi(A,B) \int_0^T Y_t^x\, dt\right] + e^{-rT}\, \mathbb{E}\left[\partial_{x_2}\Phi(A,B)\, Y_T^x\right]. \tag{12}$$

Therefore, (8) is equivalent to

$$\mathbb{E}\left[\partial_{x_1}\Phi(A,B)\, U\right] + \mathbb{E}\left[\partial_{x_2}\Phi(A,B)\, V\right] = \mathbb{E}\left[\partial_{x_1}\Phi(A,B) \int_0^T Y_t^x\, dt\right] + \mathbb{E}\left[\partial_{x_2}\Phi(A,B)\, Y_T^x\right]$$

for every $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$.

Sufficiency : Assume (9) and (10). Since $\partial_{x_1}\Phi(A,B)$ and $\partial_{x_2}\Phi(A,B)$ are $\mathcal{G}$-measurable, we get

$$\mathbb{E}\left[\partial_{x_1}\Phi(A,B)\,U\right] = \mathbb{E}\left[\partial_{x_1}\Phi(A,B)\,\mathbb{E}[U \mid \mathcal{G}]\right] = \mathbb{E}\left[\partial_{x_1}\Phi(A,B)\,\mathbb{E}\left[\int_0^T Y_t^x\,dt\middle|\mathcal{G}\right]\right]$$

$$= \mathbb{E}\left[\partial_{x_1}\Phi(A,B)\int_0^T Y_t^x\,dt\right].$$

In the same way,

$$\mathbb{E}\left[\partial_{x_2}\Phi(A,B)\,V\right] = \mathbb{E}\left[\partial_{x_2}\Phi(A,B)\,Y_T^x\right].$$

Using (11) and (12), we obtain (8).

Necessity : Assume now that

$$\Delta(x) = e^{-rT}\ E[\Phi(A,B)\Pi], \qquad \forall\Phi\in C^1\cap\ Lip(\ R^2,\ R),$$

with $\Pi = \delta(w)$.

From the computations above, we already know that

$$E[\Phi(A,B)\Pi] =\ E[\partial_{x_1}\Phi(A,B)\,U] +\ E[\partial_{x_2}\Phi(A,B)\,V],$$

where

$$U := \int_0^T Y_t^x\int_0^t \frac{\sigma(s,X_s^x)}{Y_s^x}w_s\,ds\,dt, \qquad V := Y_T^x\int_0^T \frac{\sigma(s,X_s^x)}{Y_s^x}w_s\,ds.$$

Moreover, question a) gives

$$\Delta(x) = e^{-rT}\ E\left[\partial_{x_1}\Phi(A,B)\int_0^T Y_t^x\,dt\right] + e^{-rT}\ E[\partial_{x_2}\Phi(A,B)Y_T^x].$$

Therefore, for every $\Phi\in C^1\cap\ Lip(\ R^2,\ R)$,

$$E\left[\partial_{x_1}\Phi(A,B)\left(U-\int_0^T Y_t^x\,dt\right)\right] +\ E[\partial_{x_2}\Phi(A,B)\,(V-Y_T^x)] = 0.$$

Set

$$G := \sigma(A,B), \qquad M := U-\int_0^T Y_t^x\,dt, \qquad N := V-Y_T^x.$$

Since $\partial_{x_1}\Phi(A,B)$ and $\partial_{x_2}\Phi(A,B)$ are G-measurable, taking conditional expectations with respect to G yields

$$E[\partial_{x_1}\Phi(A,B)\ E[M\mid G]] +\ E[\partial_{x_2}\Phi(A,B)\ E[N\mid G]] = 0, \qquad \forall\Phi\in C^1\cap\ Lip(\ R^2,\ R).$$

At this point, we use the same identification principle as in the one-dimensional "if and only if" criteria proved in the course for path-independent and Asian payoffs: if two G-measurable random variables $Z_1, Z_2\in L^2(G)$ satisfy

$$E[\partial_{x_1}\Phi(A,B)Z_1] +\ E[\partial_{x_2}\Phi(A,B)Z_2] = 0, \qquad \forall\Phi\in C^1\cap\ Lip(\ R^2,\ R),$$

then necessarily $Z_1 = 0$ and $Z_2 = 0$ almost surely.

Applying this with

$$Z_1 = E[M \mid G], \qquad Z_2 = E[N \mid G],$$

we obtain

$$E[M \mid G] = 0, \qquad E[N \mid G] = 0.$$

In other words,

$$E\left[\int_0^T Y_t^x \int_0^t \frac{\sigma(s, X_s^x)}{Y_s^x} w_s \, ds \, dt \;\middle|\; \sigma(A, B)\right] = E\left[\int_0^T Y_t^x \, dt \;\middle|\; \sigma(A, B)\right],$$

and

$$E\left[Y_T^x \int_0^T \frac{\sigma(s, X_s^x)}{Y_s^x} w_s \, ds \;\middle|\; \sigma(A, B)\right] = E[Y_T^x \mid \sigma(A, B)] \, .$$

These are exactly conditions we're looking for.

2.4 Question d)

From now on, we assume that

$$X_t^x = x + B_t, \qquad 0 \le t \le T,$$

that is, we work in the Bachelier model.

d.1) We want to prove that for every $\Phi \in C^1 \cap \mathrm{Lip}(\mathbb{R}^2, \mathbb{R})$,

$$\Delta(x) = e^{-rT} \, \mathbb{E}\left[\Phi\left(\int_0^T X_s^x \, ds, X_T^x\right)\left(\frac{4}{T} B_T - \frac{6}{T^2} \int_0^T s \, dB_s\right)\right]. \tag{13}$$

We now specialize to the Bachelier model. Since

$$X_t^x = x + B_t,$$

we have

$$dX_t^x = dB_t, \qquad X_0^x = x.$$

By identification with the general SDE

$$dX_t^x = b(t, X_t^x) \, dt + \sigma(t, X_t^x) \, dB_t,$$

it follows that

$$b(t, x) = 0, \qquad \sigma(t, x) = 1.$$

Moreover, since

$$X_t^x = x + B_t,$$

the derivative with respect to the initial condition is

$$Y_t^x = \frac{\partial X_t^x}{\partial x} = 1, \qquad 0 \le t \le T.$$

Therefore, conditions (9) and (10) from question c) become

$$\mathbb{E}\left[\int_0^T \int_0^t w_s\,ds\,dt \,\middle|\, \mathcal{G}\right] = \mathbb{E}[T \mid \mathcal{G}] = T, \tag{14}$$

and

$$\mathbb{E}\left[\int_0^T w_s\,ds \,\middle|\, \mathcal{G}\right] = \mathbb{E}[1 \mid \mathcal{G}] = 1, \tag{15}$$

where

$$\mathcal{G} := \sigma\left(\int_0^T X_s^x\,ds, X_T^x\right).$$

Thus, using the result from question c), it is enough to find $w \in \mathrm{dom}(\delta)$ such that

$$\int_0^T \int_0^t w_s\,ds\,dt = T \qquad \text{and} \qquad \int_0^T w_s\,ds = 1.$$

Consider the deterministic function

$$w_s := \frac{4}{T} - \frac{6}{T^2}s, \qquad 0 \le s \le T.$$

Since w is deterministic and square-integrable on $[0, T]$, we have

$$w \in L^2([0,T]) \subset L^2_{\text{prog}}(\Omega \times [0,T]),$$

hence $w \in \mathrm{dom}(\delta)$, and

$$\delta(w) = \int_0^T w_s\,dB_s$$

by the identification of the Skorohod integral with the Itô integral for adapted processes.

We first compute

$$\int_0^T w_s\,ds = \int_0^T \left(\frac{4}{T} - \frac{6}{T^2}s\right) ds = \frac{4}{T}T - \frac{6}{T^2}\frac{T^2}{2} = 4 - 3 = 1.$$

So (15) is satisfied.

Next,

$$\int_0^T \int_0^t w_s\,ds\,dt = \int_0^T (T-s)w_s\,ds$$

by Fubini's theorem. Therefore

$$\int_0^T \int_0^t w_s\,ds\,dt = \int_0^T (T-s)\left(\frac{4}{T} - \frac{6}{T^2}s\right) ds.$$

We compute separately:

$$\frac{4}{T}\int_0^T (T-s)\,ds = \frac{4}{T} \cdot \frac{T^2}{2} = 2T,$$

and

$$\frac{6}{T^2}\int_0^T s(T-s)\,ds = \frac{6}{T^2}\left(T\int_0^T s\,ds - \int_0^T s^2\,ds\right)$$

$$= \frac{6}{T^2}\left(T\frac{T^2}{2} - \frac{T^3}{3}\right) = \frac{6}{T^2}\cdot\frac{T^3}{6} = T.$$

Hence

$$\int_0^T \int_0^t w_s\,ds\,dt = 2T - T = T.$$

So (14) is also satisfied.

Since both conditions (14) and (15) are satisfied, from question c) we obtain

$$\Delta(x) = e^{-rT}\,\mathbb{E}\left[\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)\delta(w)\right].$$

It remains to compute $\delta(w)$. Since $w_s = \frac{4}{T} - \frac{6}{T^2}s$,

$$\delta(w) = \int_0^T \left(\frac{4}{T} - \frac{6}{T^2}s\right)dB_s = \frac{4}{T}\int_0^T dB_s - \frac{6}{T^2}\int_0^T s\,dB_s.$$

But

$$\int_0^T dB_s = B_T,$$

therefore

$$\delta(w) = \frac{4}{T}B_T - \frac{6}{T^2}\int_0^T s\,dB_s.$$

Substituting into the previous formula yields

$$\Delta(x) = e^{-rT}\,\mathbb{E}\left[\Phi\left(\int_0^T X_s^x\,ds, X_T^x\right)\left(\frac{4}{T}B_T - \frac{6}{T^2}\int_0^T s\,dB_s\right)\right].$$

This is exactly (13).

d.2) We now determine whether the weight found in question d.1 is optimal in the sense of question b).

Let

$$A := \int_0^T X_s^x\,ds, \qquad B := X_T^x, \qquad G := \sigma(A, B).$$

From question b), if $\Pi \in W$, then the optimal weight is given by

$$\Pi_0 = \mathbb{E}[\Pi \mid G].$$

Therefore, it is enough to check whether the weight obtained in question d.1 is already G-measurable.

From question d.1, we found

$$\Pi = \frac{4}{T}B_T - \frac{6}{T^2}\int_0^T s\,dB_s.$$

We now rewrite this quantity in terms of A and B.

Since, in the Bachelier model,

$$X_t^x = x + B_t,$$

we have

$$B = X_T^x = x + B_T, \qquad \text{hence} \qquad B_T = B - x.$$

Moreover,

$$A = \int_0^T X_s^x\,ds = \int_0^T (x + B_s)\,ds = xT + \int_0^T B_s\,ds,$$

so that

$$\int_0^T B_s\,ds = A - xT.$$

Next, using integration by parts for sB_s, we get

$$d(sB_s) = B_s\,ds + s\,dB_s.$$

Integrating over $[0,T]$ yields

$$TB_T = \int_0^T B_s\,ds + \int_0^T s\,dB_s,$$

and therefore

$$\int_0^T s\,dB_s = TB_T - \int_0^T B_s\,ds.$$

Substituting this into the expression of Π, we obtain

$$\Pi = \frac{4}{T}B_T - \frac{6}{T^2}\left(TB_T - \int_0^T B_s\,ds\right) = -\frac{2}{T}B_T + \frac{6}{T^2}\int_0^T B_s\,ds.$$

Using the identities $B_T = B - x$ and $\int_0^T B_s\,ds = A - xT$, this becomes

$$\Pi = -\frac{2}{T}(B - x) + \frac{6}{T^2}(A - xT) = \frac{6}{T^2}A - \frac{2}{T}B - \frac{4x}{T}.$$

Hence, Π is an affine deterministic function of (A,B). In particular, Π is G-measurable. Therefore,

$$\mathbb{E}[\Pi \mid G] = \Pi.$$

From the result of question b), this shows that Π coincides with the optimal weight.

We conclude that the weight found in question d.1 is optimal:

$$\boxed{\Pi = \frac{4}{T}B_T - \frac{6}{T^2}\int_0^T s\,dB_s}$$

is the optimal weight in the sense of question b).

3 Part C : Numerical Part

In this part, we implement the Monte Carlo methods. We work in the Bachelier model

$$X_t^x = x + B_t,$$

with parameters

$$x = 100, \qquad r = 10\%, \qquad T = 1,$$

and payoff

$$\Phi\left(\int_0^T X_t^x \, dt, \; X_T^x\right) = \mathbf{1}_{\left\{\int_0^T X_t^x \, dt < 110\right\}} (X_T^x - 100)_+.$$

Since the payoff depends on the time integral of the process, we approximate

$$\int_0^T X_t^x \, dt$$

by the discrete quantity

$$\frac{T}{M} \sum_{k=0}^{M} X_{kT/M}^x,$$

We will compare the results obtained for

$$M = 50, \qquad M = 150, \qquad M = 250.$$

For each method, we report the Monte Carlo estimator, its empirical variance, and a confidence interval. We also study the effect of the number of simulations N, and, for the finite-difference method, the effect of the perturbation parameter ε. Finally, we compare the finite-difference estimator with the Malliavin weight estimator in terms of stability and empirical variance.

3.1 Question a)

We estimate the price of the option by Monte Carlo simulation in the Bachelier model

$$X_t^x = x + B_t, \qquad x = 100, \quad r = 10\%, \quad T = 1.$$

The payoff is

$$\Phi\left(\int_0^T X_t^x \, dt, \; X_T^x\right) = \mathbf{1}_{\left\{\int_0^T X_t^x \, dt < 110\right\}} (X_T^x - 100)_+.$$

Since the payoff depends on the time integral of the process, we use the approximation required in the statement:

$$\int_0^T X_t^x \, dt \approx \frac{T}{M} \sum_{k=0}^{M} X_{kT/M}^x,$$

with

$$M \in \{50, 150, 250\}.$$

For a fixed value of M, the Monte Carlo estimator of the price is

$$\widehat{P}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{\widehat{A}^{(i)} < 110\}} \left(X_T^{x,(i)} - 100\right)_+,$$

where

$$\widehat{A}^{(i)} = \frac{T}{M} \sum_{k=0}^{M} X_{kT/M}^{x,(i)}.$$

We also compute the empirical variance of the discounted payoff, the empirical variance of the Monte Carlo estimator, and a 95% confidence interval based on the central limit theorem.

Final numerical estimates Table 1 reports the final Monte Carlo estimates for the three values of M, with $N = 51000$.

Table 1: Monte Carlo price estimation for the complex Asian option.

M	N	$\widehat{P}_N$	Sample variance	Estimator variance	95% CI
50	51000	0.364212	0.284251	5.57×10^{-6}	[0.359584, 0.368839]
150	51000	0.363835	0.283481	5.56×10^{-6}	[0.359214, 0.368456]
250	51000	0.363415	0.283415	5.56×10^{-6}	[0.358795, 0.368036]

The three estimates are very close, which suggests that the time discretization error is already quite small for $M = 50$, and becomes negligible compared with the Monte Carlo error when M increases to 150 and 250. In particular, the values obtained for $M = 150$ and $M = 250$ are almost identical.

Empirical convergence with respect to N We now study the empirical convergence of the price estimator as the number of simulations increases.

Figure 1 shows that the estimator stabilizes as N increases. For small values of N, the fluctuations are relatively important, which is typical of Monte Carlo methods. As N becomes larger, the three curves become much more stable and remain in a narrow band around 0.364. This behavior is consistent with the law of large numbers.

Variance analysis To quantify the statistical uncertainty of the estimator, we also represent the empirical variance of the Monte Carlo estimator as a function of N.

Figure 2 confirms that the variance of the estimator decreases as N increases. More precisely, the decay is consistent with the usual Monte Carlo rate, since the variance of the estimator is of order $1/N$. Moreover, the three curves are almost indistinguishable, which again indicates that the choice of M has only a limited impact on the final price estimate, at least for the values considered here.

Example of intermediate result For instance, with $N = 5000$ and $M = 50$, we obtained

$$\widehat{P}_N = 0.361023, \qquad \widehat{\mathrm{Var}}(\Phi) = 0.284009, \qquad \widehat{\mathrm{Var}}(\widehat{P}_N) = 5.68 \times 10^{-5},$$

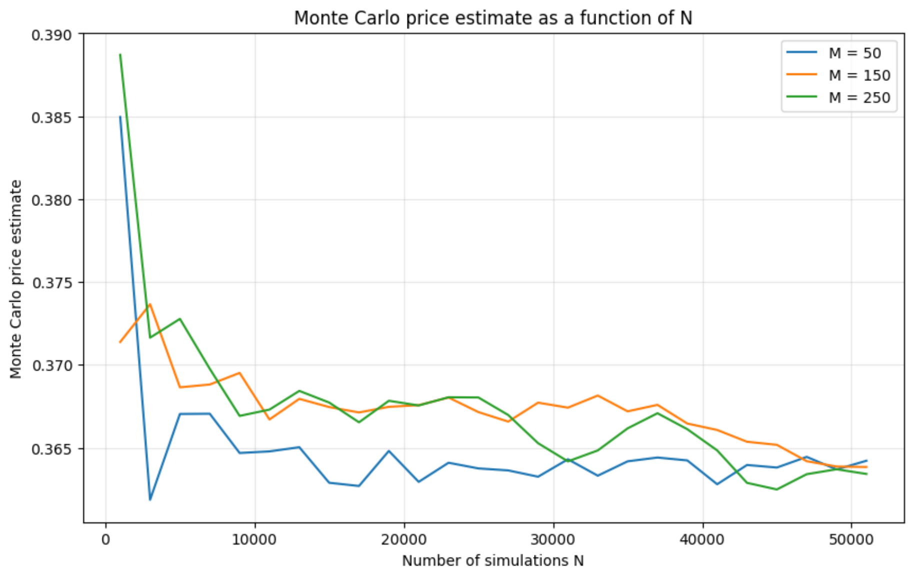

Figure 1: Monte Carlo price estimate as a function of N, for $M = 50, 150, 250$.

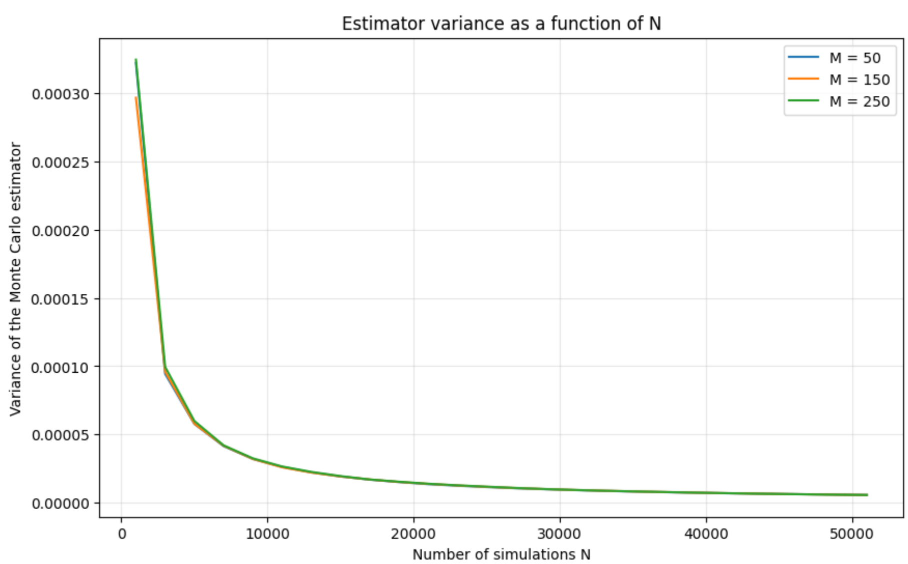

Figure 2: Variance of the Monte Carlo estimator as a function of N, for $M = 50, 150, 250$.

with standard error

$$0.007537$$

and 95% confidence interval

$$[0.346252,\ 0.375795].$$

This intermediate result is consistent with the final values and illustrates the progressive reduction of statistical uncertainty as N increases.

Overall, the Monte Carlo price estimator appears stable for large values of N, and the numerical results suggest that the discretization error induced by the approximation of the time integral is relatively small compared with the Monte Carlo error for the values $M = 50, 150, 250$.

3.2 Question b)

We now estimate the Delta of the option by a finite-difference method. We use the centered estimator

$$\widehat{\Delta}^{FD}_{N,\varepsilon} = \frac{\widehat{P}_N(x+\varepsilon) - \widehat{P}_N(x-\varepsilon)}{2\varepsilon},$$

where $\widehat{P}_N(x+\varepsilon)$ and $\widehat{P}_N(x-\varepsilon)$ are the Monte Carlo estimators of the price corresponding to the perturbed initial values $x+\varepsilon$ and $x-\varepsilon$. In the implementation, the same Brownian paths are used for both terms in order to reduce the variance of the estimator.

As in question a), we report the empirical variance of the sample, the empirical variance of the Monte Carlo estimator, and a 95% confidence interval based on the central limit theorem.

Intermediate result As a first illustration, for

$$N = 5000, \qquad M = 50, \qquad \varepsilon = 1,$$

we obtained

$$\widehat{\Delta}^{FD}_{N,\varepsilon} = 0.453506,$$

with sample variance

$$0.104523,$$

estimator variance

$$2.09 \times 10^{-5},$$

standard error

$$0.004572,$$

and 95% confidence interval

$$[0.444544,\ 0.462467].$$

This already gives a reasonable estimate of the Delta, although the confidence interval is still relatively wide.

Final estimates for $\varepsilon = 1$. We first compare the results obtained for $M = 50, 150, 250$ when $\varepsilon = 1$ and $N = 51000$.

The three estimates are very close. This suggests that, for the values of M considered here, the discretization error has only a limited effect on the finite-difference estimator of the Delta. In particular, increasing M from 150 to 250 does not significantly modify the result.

Table 2: Finite-difference Delta estimation for $\varepsilon = 1$ and $N = 51000$.

M	$\widehat{\Delta}^{FD}$	Sample variance	Estimator variance	Standard error	95% CI
50	0.452692	0.106168	2.0817×10^{-6}	0.001443	[0.449864, 0.455520]
150	0.452840	0.105979	2.0780×10^{-6}	0.001442	[0.450015, 0.455665]
250	0.453044	0.105718	2.0729×10^{-6}	0.001440	[0.450223, 0.455866]

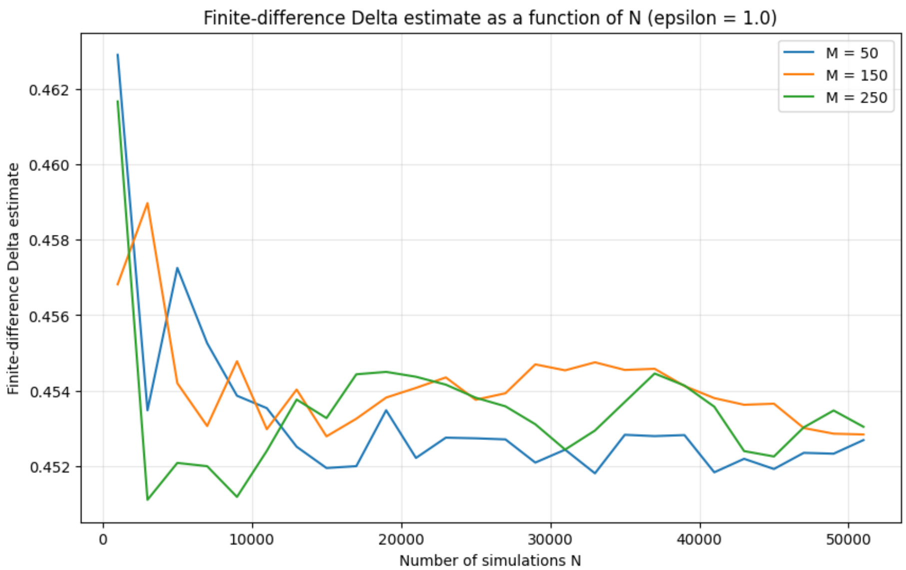

Figure 3: Finite-difference Delta estimate as a function of N, for $\varepsilon = 1$ and $M = 50, 150, 250$.

Convergence with respect to N We now study the behavior of the estimator as the number of simulations increases.

Figure 3 shows that the estimator stabilizes as N increases. For small values of N, the fluctuations are more visible, while for large values of N the estimates remain in a narrow band around 0.453. This is consistent with the usual behavior of Monte Carlo estimators.

Figure 4 confirms that the variance of the estimator decreases as N increases. Moreover, the three curves are almost indistinguishable, which again indicates that the choice of M has only a limited impact on the statistical precision of the estimator in this experiment.

Effect of the perturbation parameter ε We now investigate the effect of ε on the estimator. Table 3 reports the results obtained for $M = 150$ and $N = 51000$.

The values of $\widehat{\Delta}^{FD}$ remain very close for all tested values of ε, which indicates that the point estimate itself is rather stable. However, the variance of the estimator decreases significantly when ε increases. The same phenomenon is observed for the width of the confidence interval.

Figure 5 shows that the delta estimate varies only slightly with ε. By contrast, Figures 6 and 7 show that the statistical uncertainty decreases markedly as ε increases. In our numerical experiments, small values of ε lead to a noisier estimator, while larger values of ε produce a much

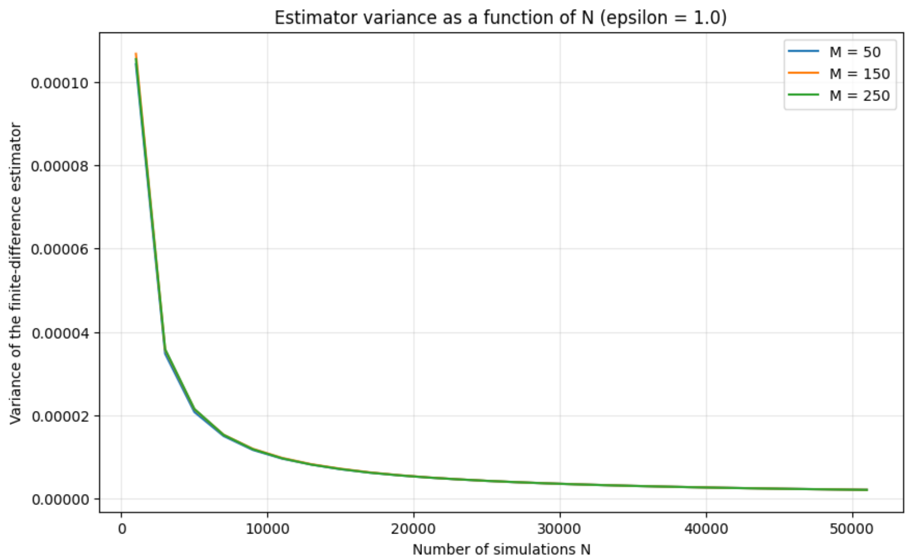

Figure 4: Variance of the finite-difference estimator as a function of N, for $\varepsilon = 1$ and $M = 50, 150, 250$.

Table 3: Effect of ε on the finite-difference estimator for $M = 150$ and $N = 51000$.

ε	$\widehat{\Delta}^{FD}$	Sample variance	Estimator variance	Standard error	95% CI
0.1	0.452461	0.193704	3.7981×10^{-6}	0.001949	[0.448641, 0.456281]
0.25	0.452619	0.177648	3.4833×10^{-6}	0.001866	[0.448961, 0.456277]
0.5	0.452909	0.151611	2.9728×10^{-6}	0.001724	[0.449530, 0.456288]
1.0	0.452840	0.105979	2.0780×10^{-6}	0.001442	[0.450015, 0.455665]
2.0	0.452752	0.047709	9.3547×10^{-7}	0.000967	[0.450857, 0.454648]

more stable estimate.

Overall, the finite-difference method provides a consistent estimate of the delta, and the estimator clearly stabilizes as N increases. The choice of the time discretization parameter M has little influence on the final result, whereas the choice of ε has a significant impact on the variance and on the width of the confidence interval. This illustrates the usual bias-variance trade-off of finite-difference methods.

3.3 Question c)

We now estimate the Delta using the Malliavin integration by parts formula obtained in part b with question d.1). In the Bachelier model, we have

$$\Delta(x) = e^{-rT}\,\mathbb{E}\left[\Phi\left(\int_0^T X_t^x\,dt,\ X_T^x\right)\Pi\right],$$

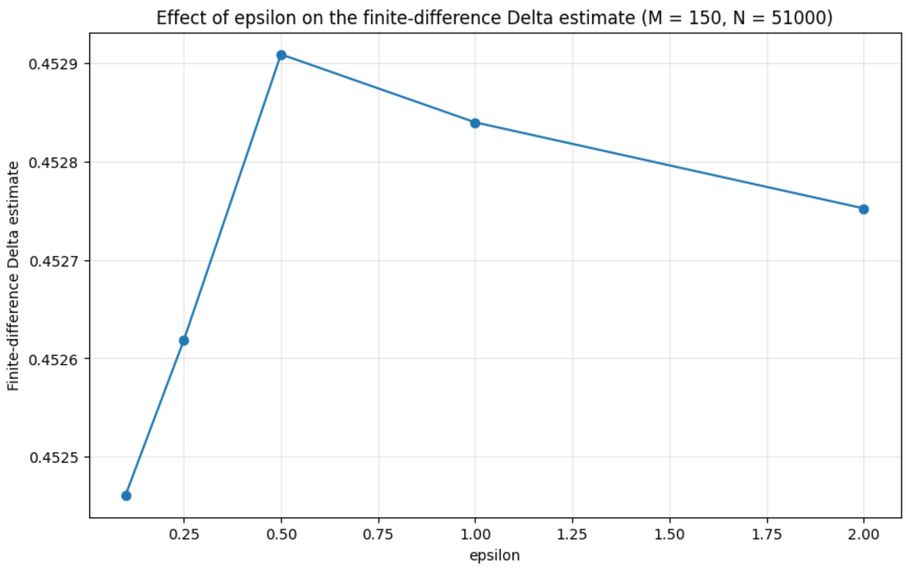

Figure 5: Effect of ε on the finite-difference Delta estimate, for $M = 150$ and $N = 51000$.

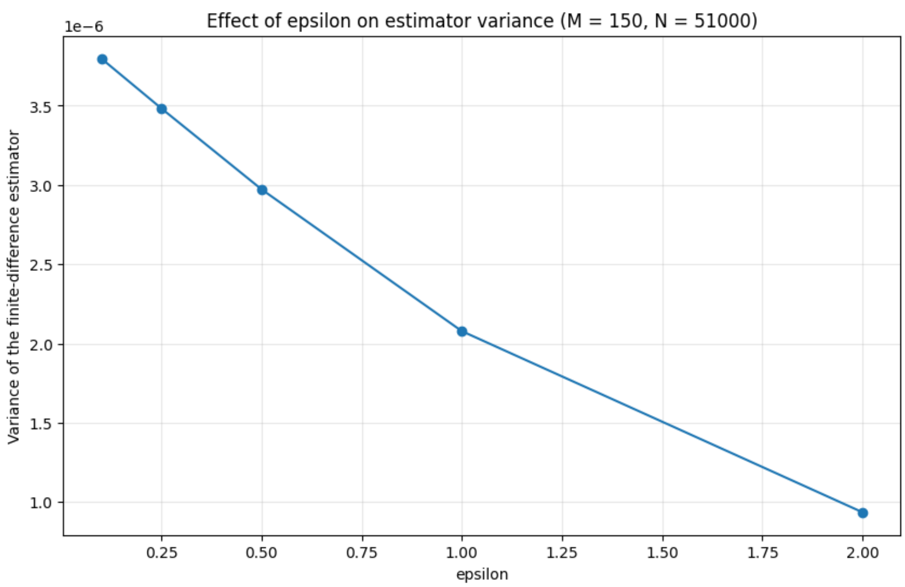

Figure 6: Effect of ε on the variance of the finite-difference estimator, for $M = 150$ and $N = 51000$.

where

$$\Pi = \frac{4}{T} B_T - \frac{6}{T^2} \int_0^T s \, dB_s.$$

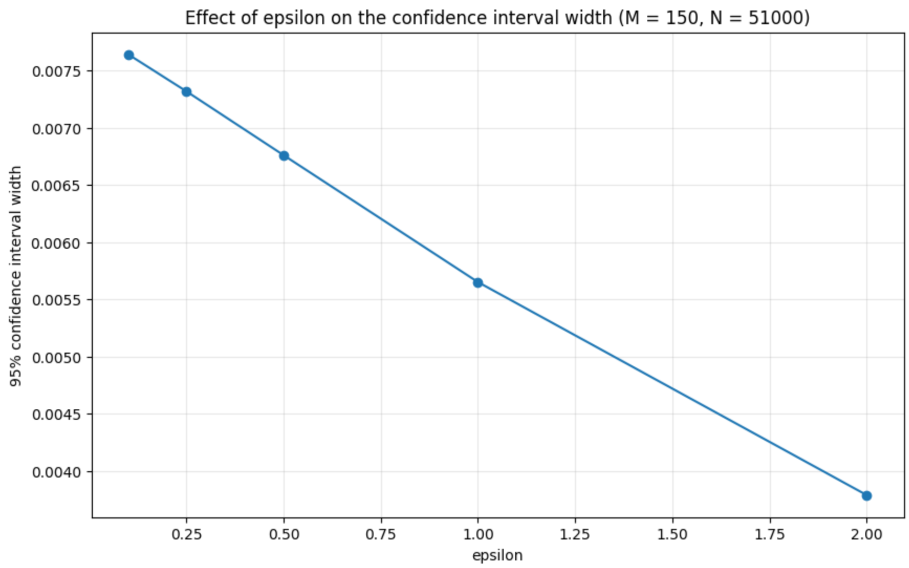

Figure 7: Effect of ε on the width of the 95% confidence interval, for $M = 150$ and $N = 51000$.

Therefore, the corresponding Monte Carlo estimator is

$$\widehat{\Delta}_N^{Mal} = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \Phi\Big(\widehat{A}^{(i)}, X_T^{x,(i)}\Big) \, \Pi^{(i)},$$

with

$$\widehat{A}^{(i)} = \frac{T}{M} \sum_{k=0}^{M} X_{kT/M}^{x,(i)}.$$

As in the previous questions, we report the empirical variance of the simulated sample, the empirical variance of the Monte Carlo estimator, and a 95% confidence interval based on the central limit theorem.

Intermediate result For the configuration

$$N = 5000, \qquad M = 50,$$

we obtained

$$\widehat{\Delta}_N^{Mal} = 0.486683,$$

with sample variance

$$2.476331,$$

estimator variance

$$4.9527 \times 10^{-4},$$

standard error

$$0.022255,$$

and 95% confidence interval

$$[0.443065,\ 0.530301].$$

Compared with the finite-difference estimator, this first result already suggests that the Malliavin estimator has a much larger variance. The confidence interval is also significantly wider, which indicates a higher statistical uncertainty for a comparable number of simulations.

Convergence with respect to N We now study the behavior of the Malliavin estimator as the number of simulations increases.

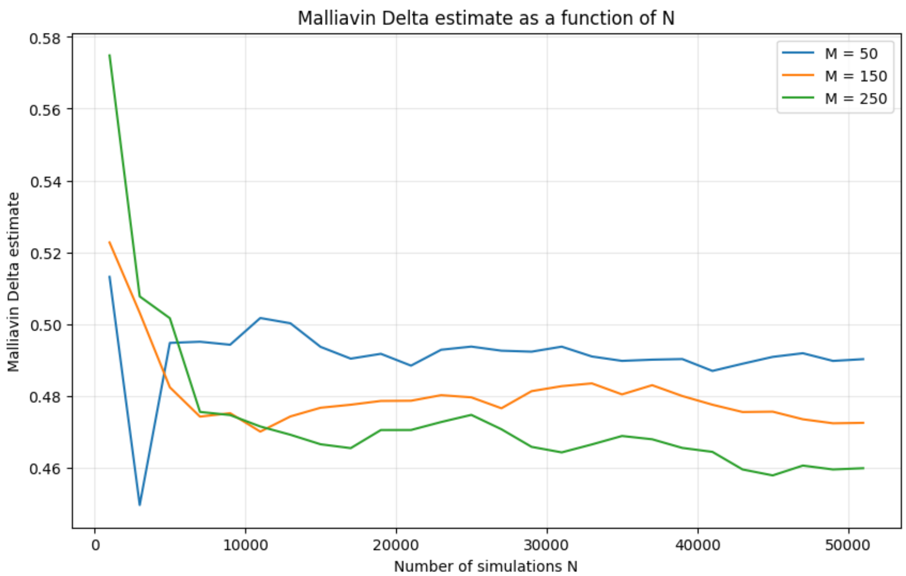

Figure 8: Malliavin Delta estimate as a function of N, for $M = 50, 150, 250$.

Figure 8 shows that the estimator exhibits stronger fluctuations than in the finite-difference case, especially for small and moderate values of N. Although the curves tend to stabilize as N increases, the convergence appears slower and noisier. We also observe visible differences between the three values of M, which suggests that the discretization of the stochastic integral entering the Malliavin weight has a non-negligible impact on the numerical result.

Confidence interval width To quantify the statistical uncertainty of the estimator, we also plot the width of the 95% confidence interval as a function of N.

Figure 9 confirms that the confidence interval width decreases when N increases, as expected for a Monte Carlo estimator. However, the interval remains relatively large even for the largest tested values of N, reflecting the important variance of the Malliavin estimator in this example.

Overall, the Malliavin method provides a valid estimator of the Delta, but in this numerical experiment it appears substantially more variable than the finite-difference estimator. While the

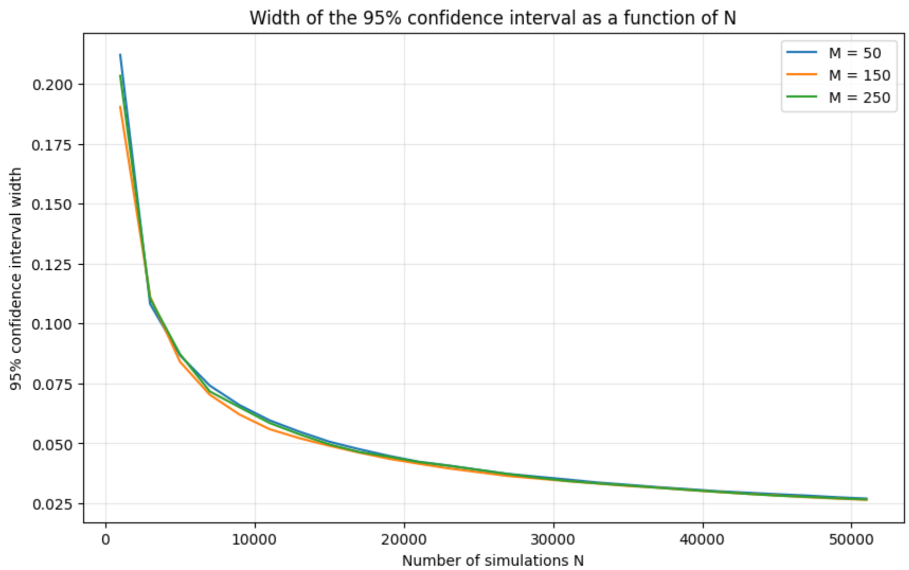

Figure 9: Width of the 95% confidence interval as a function of N, for $M = 50, 150, 250$.

confidence interval width decreases with N, the estimator remains noticeably noisy, which suggests that the Malliavin weight derived in part B with question d.1) is not numerically competitive here, despite its theoretical interest.

3.4 Question d)

We now compare the two Monte Carlo estimators of the Delta:

- the centered finite-difference estimator with $\varepsilon = 1$,
- the Malliavin weight estimator obtained in Question c).

The comparison is carried out for

$$M \in \{50, 150, 250\}, \qquad N = 51000,$$

and is based on the Delta estimates, the empirical variance of the estimators, and the width of the 95% confidence intervals.

Final comparison for $N = 51000$ Table 4 summarizes the results obtained for the two methods. The first observation is that the finite-difference estimates are very stable with respect to M, since all values lie in a very narrow interval around 0.453. By contrast, the Malliavin estimates vary much more with the discretization level: they decrease from about 0.490 for $M = 50$ to about 0.460 for $M = 250$. This suggests that the Malliavin estimator is more sensitive to the time discretization used in the numerical implementation.

Table 4: Comparison between finite differences and the Malliavin estimator for $N = 51000$.

M	$\widehat{\Delta}^{FD}$	$\widehat{\Delta}^{Mal}$	$\widehat{\mathrm{Var}}(\widehat{\Delta}^{FD})$	$\widehat{\mathrm{Var}}(\widehat{\Delta}^{Mal})$	Ratio Var(Mal/FD)	Ratio CI width(Mal/FD)
50	0.452692	0.490259	2.08×10^{-6}	4.74×10^{-5}	22.767152	4.771494
150	0.452840	0.472527	2.08×10^{-6}	4.56×10^{-5}	21.939644	4.683977
250	0.453044	0.459891	2.07×10^{-6}	4.61×10^{-5}	22.256118	4.717639

The second and most important observation concerns the variance. For all values of M, the empirical variance of the Malliavin estimator is approximately 22 times larger than the variance of the finite-difference estimator. As a consequence, the 95% confidence intervals are also much wider for the Malliavin method, by a factor close to 4.7. Hence, in this numerical experiment, the finite-difference estimator is clearly more accurate from a statistical point of view.

Comparison of the Delta estimates as a function of N We now compare the two methods graphically as the number of simulations increases.

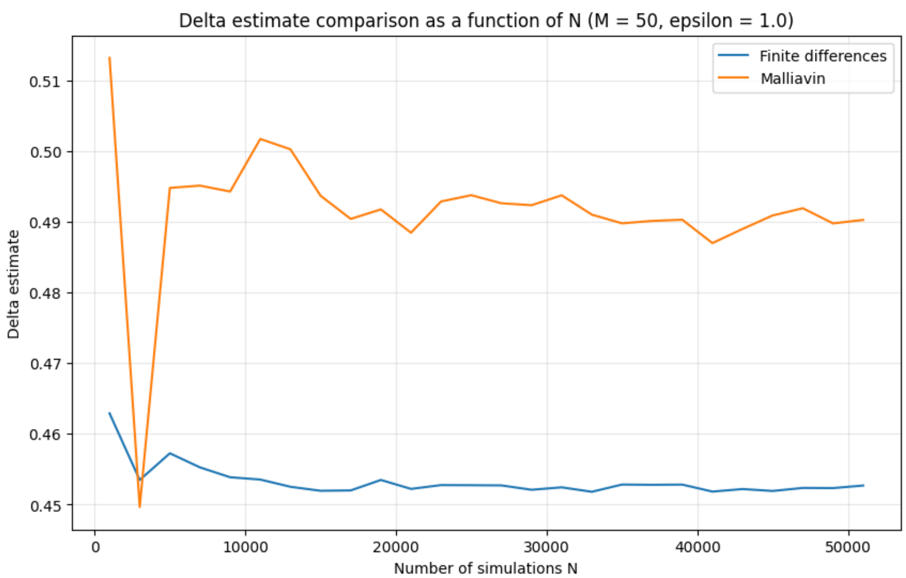

Figure 10: Comparison of the Delta estimates as a function of N, for $M = 50$ and $\varepsilon = 1$.

Figures 10–12 show that the finite-difference estimator stabilizes rapidly around a value close to 0.453, with relatively small fluctuations. The Malliavin estimator is much noisier, especially for small and moderate values of N, and its limiting level appears to depend more strongly on M. This confirms the conclusion drawn from Table 4.

Comparison of the estimator variances We now compare the empirical variances of the two estimators.

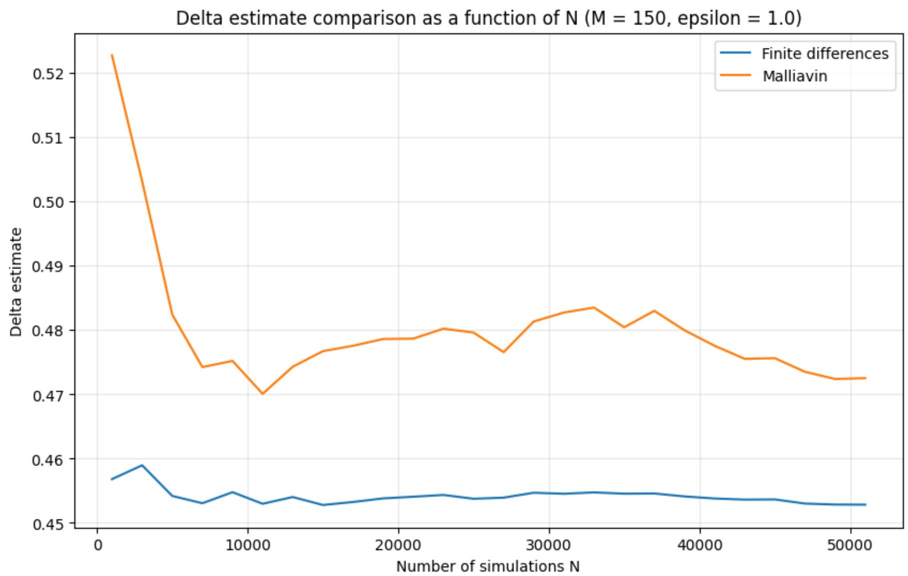

Figure 11: Comparison of the Delta estimates as a function of N, for $M = 150$ and $\varepsilon = 1$.

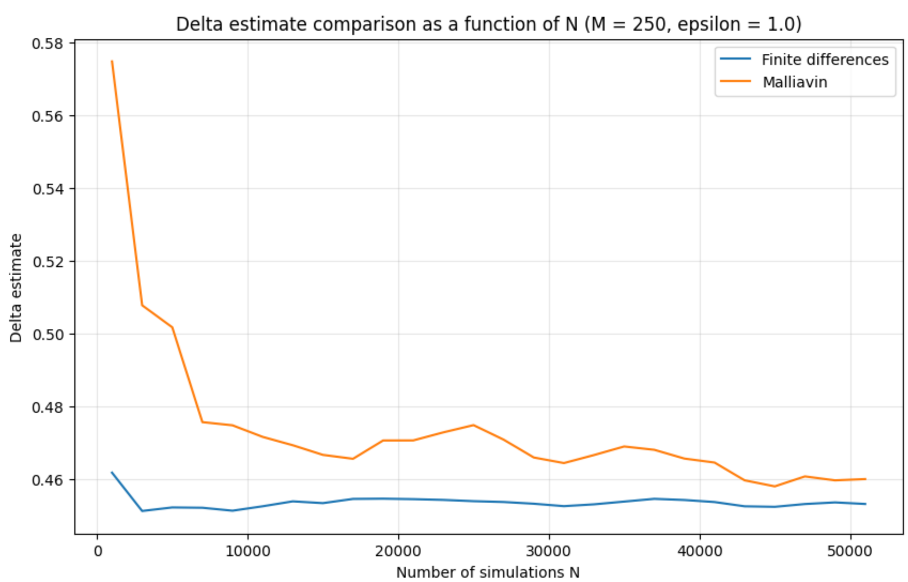

Figure 12: Comparison of the Delta estimates as a function of N, for $M = 250$ and $\varepsilon = 1$.

Figures 13–15 clearly show that the Malliavin estimator has a much larger variance than the finite-difference estimator for all tested values of N. Although both variances decrease when N increases, the gap between the two methods remains very large throughout the numerical experiment.

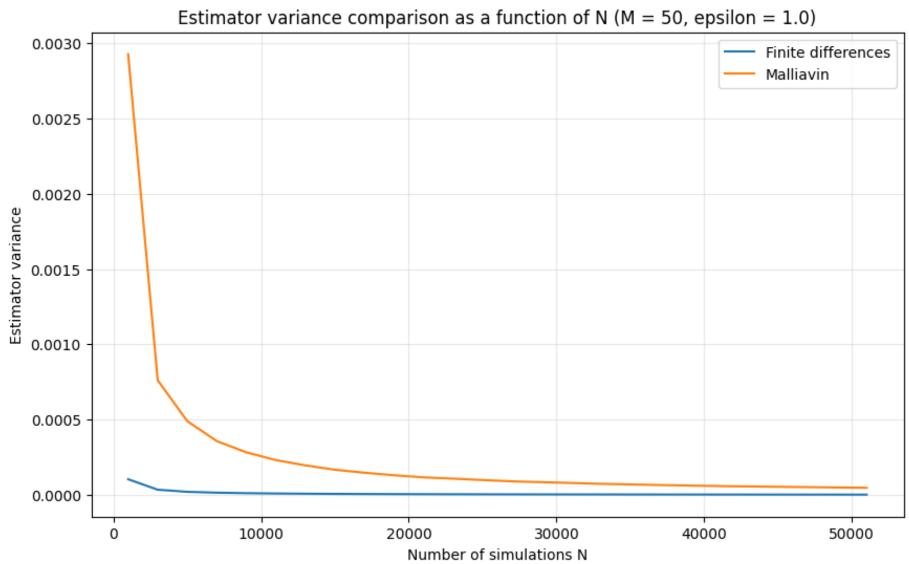

Figure 13: Comparison of the estimator variances as a function of N, for $M = 50$ and $\varepsilon = 1$.

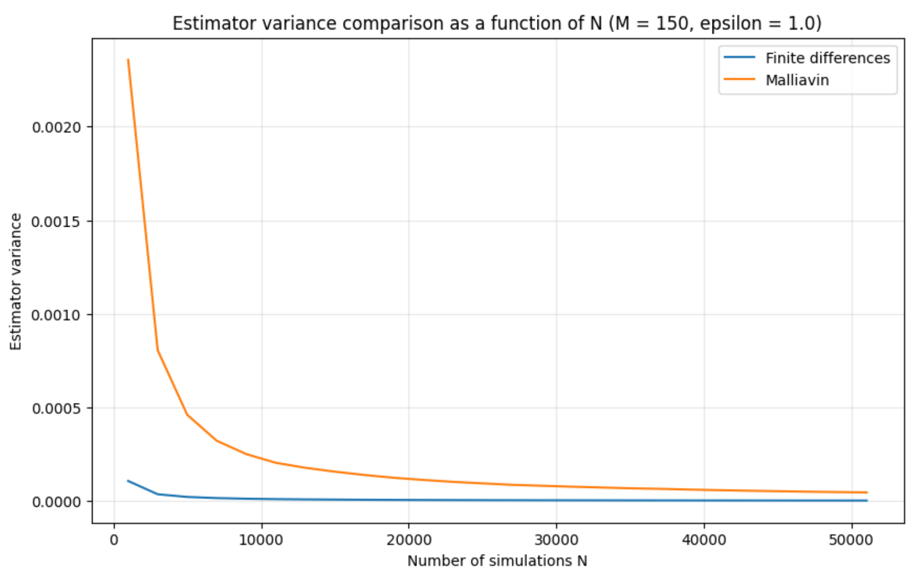

Figure 14: Comparison of the estimator variances as a function of N, for $M = 150$ and $\varepsilon = 1$.

Comparison of the confidence interval widths Finally, we compare the widths of the 95% confidence intervals.

Figures 16–18 lead to the same conclusion: the confidence intervals associated with the Malliavin estimator are much wider than those obtained with finite differences. Thus, for a fixed simulation

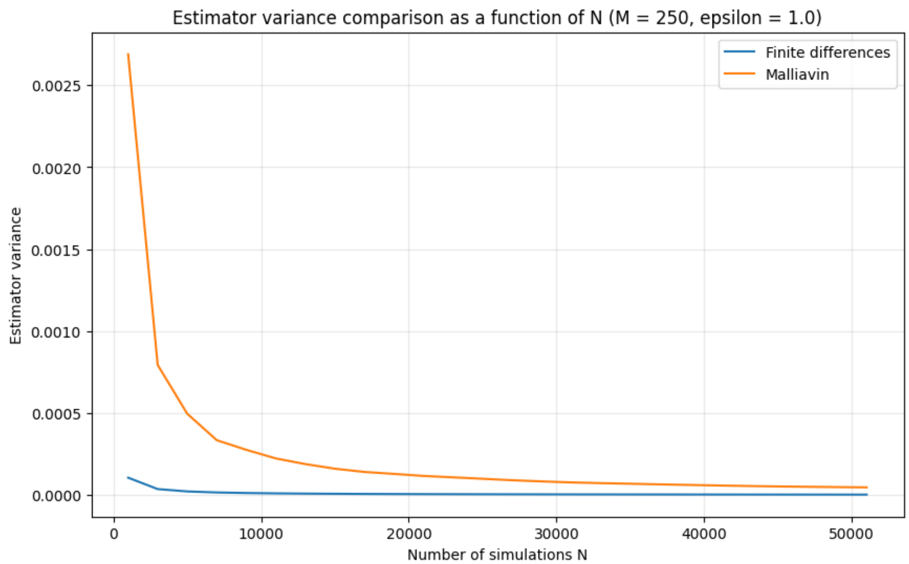

Figure 15: Comparison of the estimator variances as a function of N, for $M = 250$ and $\varepsilon = 1$.

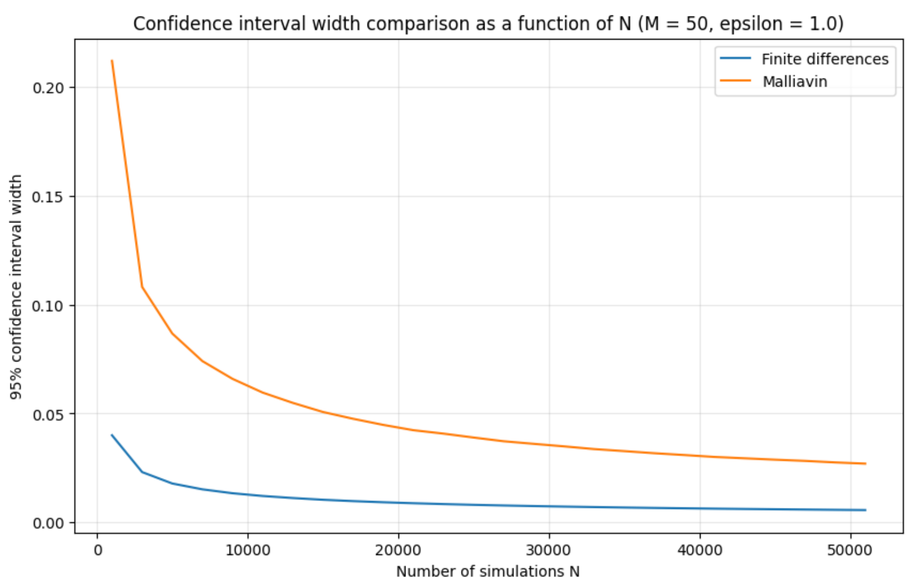

Figure 16: Comparison of the 95% confidence interval widths as a function of N, for $M = 50$ and $\varepsilon = 1$.

budget, the finite-difference method provides a much more precise estimate of the Delta in the present setting.

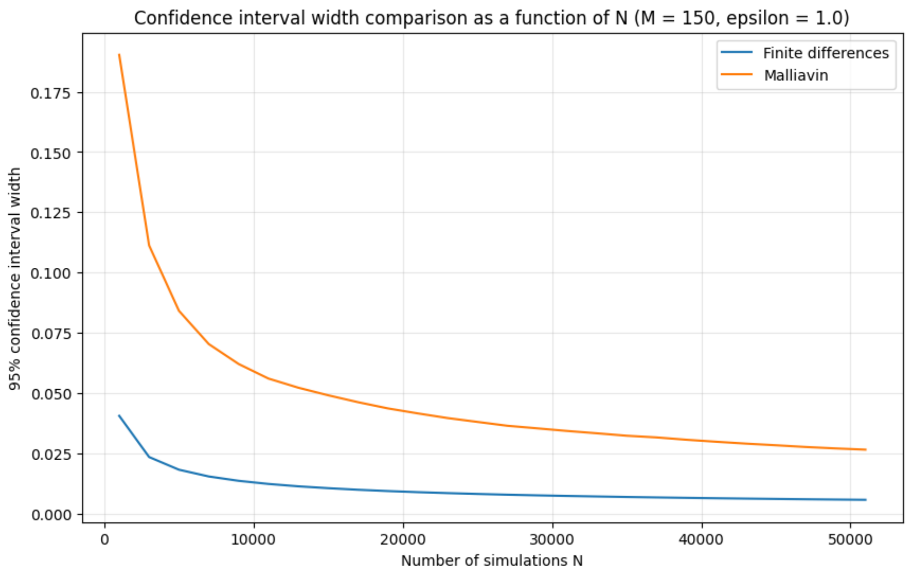

Figure 17: Comparison of the 95% confidence interval widths as a function of N, for $M = 150$ and $\varepsilon = 1$.

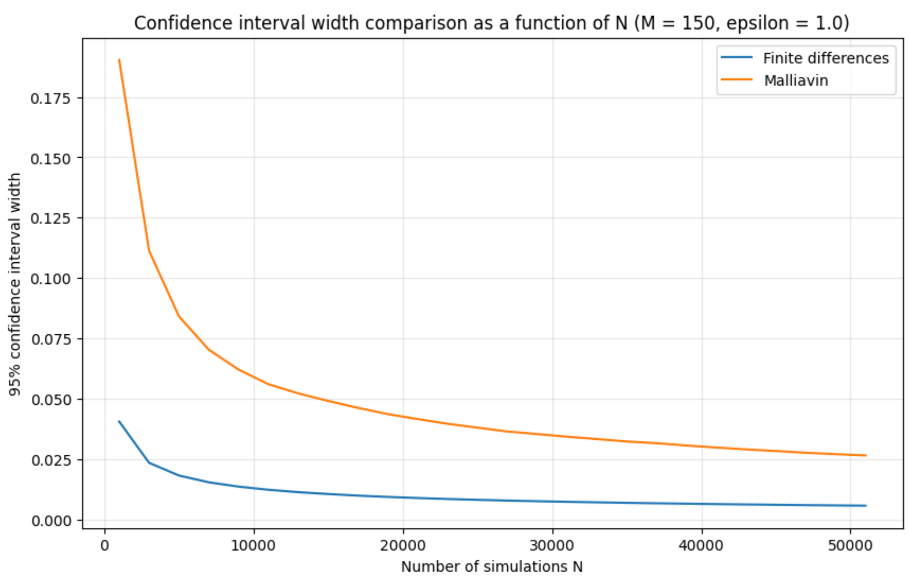

Figure 18: Comparison of the 95% confidence interval widths as a function of N, for $M = 250$ and $\varepsilon = 1$.

Conclusion Overall, the numerical comparison strongly favors the finite-difference estimator. In our implementation, it is more stable with respect to the discretization parameter M, has a much smaller empirical variance, and produces significantly narrower confidence intervals. The Malliavin estimator remains theoretically valid, but it is much noisier in this example. Therefore, for this payoff and for the chosen numerical procedure, the finite-difference method appears to be the most efficient approach for estimating the Delta.

3.5 Question e)

The numerical experiments carried out in questions a) to d) allow us to draw several conclusions.
First, concerning the price estimation, the Monte Carlo estimator is stable and behaves as expected when the number of simulations N increases. The empirical variance decreases regularly, in accordance with the usual Monte Carlo rate, and the three discretization levels $M = 50$, $M = 150$, and $M = 250$ lead to very similar results. This suggests that, for the present payoff, the discretization error induced by the approximation of the time integral is relatively small compared with the statistical error.
Second, for the estimation of the Delta by finite differences, the method provides very stable results. The estimator depends only weakly on the time discretization parameter M, and the Delta estimates remain close to 0.453 in all our experiments. The main sensitivity comes from the perturbation parameter ε: while the point estimate remains quite stable, the variance and the width of the confidence interval decrease significantly when ε increases. This illustrates the usual bias-variance trade-off of finite-difference methods.
Third, the Malliavin weight estimator is theoretically valid, but in our numerical implementation it turns out to be much noisier. Its fluctuations are larger, its confidence intervals are much wider, and its values appear to be more sensitive to the discretization parameter M. In particular, even for large values of N, the empirical variance remains much larger than in the finite-difference case.
Finally, the direct comparison of the two methods clearly favors the finite-difference estimator. For all tested values of M, the empirical variance of the Malliavin estimator is approximately 22 times larger than that of the finite-difference estimator, and the corresponding confidence intervals are about 4.7 times wider. Therefore, for this payoff and for the numerical procedures used here, the finite-difference method is more efficient and more accurate from a statistical point of view.
Overall, the Monte Carlo study shows that the price is estimated reliably and that the finite-difference method provides the best numerical performance for the Delta in this setting. Although the Malliavin approach is theoretically appealing, it is not numerically competitive here. More precisely, our results do not reproduce the favorable variance reduction effect of the Malliavin method that was illustrated in the course for some irregular payoffs. In the present framework, this difference may be explained by the specific features of the experiment, in particular the Bachelier model, the discretization of the stochastic integral appearing in the Malliavin weight, and the use of a centered finite-difference estimator with common random numbers.

4 Use of artificial intelligence

Artificial intelligence was used in this project as a support tool rather than as a substitute for personal work. On the theoretical side, it was mainly helpful for clarifying some points of Malliavin calculus, identifying which results from the lecture notes were relevant for each question, and improving the overall structure of some proofs and arguments. In particular, it helped me reformulate certain parts more clearly and detect steps that required a stronger justification. However, I remained cautious with these contributions, since an argument that looks mathematically convincing may still rely on unjustified assumptions or shortcuts. For this reason, every theoretical suggestion had to be checked again against the course material and the logic of the proof. On the numerical side, AI was useful for discussing the implementation of the Monte Carlo methods, verifying the coherence of the Python code, and helping interpret the numerical results obtained through the tables and graphs. It also helped me compare the finite-difference and Malliavin approaches in a more structured way. That being said, the final responsibility remained mine: the formulas had to be verified, the code had to be tested, and the conclusions had to be drawn from the actual outputs rather than from automatically generated comments. Overall, AI was valuable as a tool for clarification, organization, and verification, but it could not replace either mathematical rigor or critical interpretation.